

# 1. View Authentication Plugin of Users

```
SELECT user, host, plugin  
FROM mysql.user;
```

Example Output:

| user     | host      | plugin                |
|----------|-----------|-----------------------|
| root     | localhost | auth_socket           |
| app_user | %         | caching_sha2_password |

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## 2. Create User Using caching\_sha2\_password

(Default in MySQL 8)

```
CREATE USER 'data_engineer'@'localhost'  
IDENTIFIED WITH caching_sha2_password  
BY 'StrongPass@123';
```

Grant privileges:

```
GRANT SELECT, INSERT, UPDATE  
ON company_db.*  
TO 'data_engineer'@'localhost';
```

---

## 3. Create User Using mysql\_native\_password

Older applications sometimes require this plugin.

```
CREATE USER 'legacy_app'@'%'  
IDENTIFIED WITH mysql_native_password
```

```
BY 'Legacy@123';
```

Grant privileges:

```
GRANT ALL PRIVILEGES  
ON sales_db.*  
TO 'legacy_app'@'%';
```

---

## 4. Change Existing User Authentication Plugin

Example:

```
ALTER USER 'legacy_app'@'%'  
IDENTIFIED WITH caching_sha2_password  
BY 'NewStrongPass@123';
```

---

## 5. Create User with sha256\_password

More secure than older native password method.

```
CREATE USER 'secure_user'@'localhost'  
IDENTIFIED WITH sha256_password  
BY 'Secure@456';
```

---

## 6. Linux Socket Authentication (auth\_socket)

Used mainly for local Linux system users.

User can log in without password if OS username matches MySQL username.

```
CREATE USER 'ubuntu_user'@'localhost'  
IDENTIFIED WITH auth_socket;
```

Login:

```
sudo mysql -u ubuntu_user
```

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## 7. Check Available Authentication Plugins

```
SHOW PLUGINS;
```

Filter authentication plugins:

```
SELECT PLUGIN_NAME, PLUGIN_STATUS  
FROM INFORMATION_SCHEMA.PLUGINS  
WHERE PLUGIN_TYPE = 'AUTHENTICATION';
```

---

## 8. Remove User

```
DROP USER 'legacy_app'@'%';
```

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## 9. Verify User Permissions

```
SHOW GRANTS FOR 'data_engineer'@'localhost';
```

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## 10. Example Enterprise Security Setup

### Reporting User

```
CREATE USER 'report_user'@'10.10.%'  
IDENTIFIED WITH caching_sha2_password  
BY 'Report@123';
```

```
GRANT SELECT  
ON reporting_db.*  
TO 'report_user'@'10.10.%';
```

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# Read-Only Auditor

```
CREATE USER 'auditor'@'localhost'  
IDENTIFIED WITH sha256_password  
BY 'Audit@123';
```

```
GRANT SELECT, SHOW VIEW  
ON *.*  
TO 'auditor'@'localhost';
```

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# Important Interview Points

| Plugin                | Usage                      | Security          |
|-----------------------|----------------------------|-------------------|
| caching_sha2_password | Default in MySQL 8         | High              |
| mysql_native_password | Legacy support             | Medium            |
| sha256_password       | Strong authentication      | High              |
| auth_socket           | Linux local authentication | High (local only) |

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# Best Practices

- Prefer `caching_sha2_password` in MySQL 8+
- Avoid `%` host wildcard unless required
- Use strong passwords
- Rotate credentials regularly
- Separate application users and admin users
- Grant minimum privileges only
- Use SSL/TLS for remote DB connections